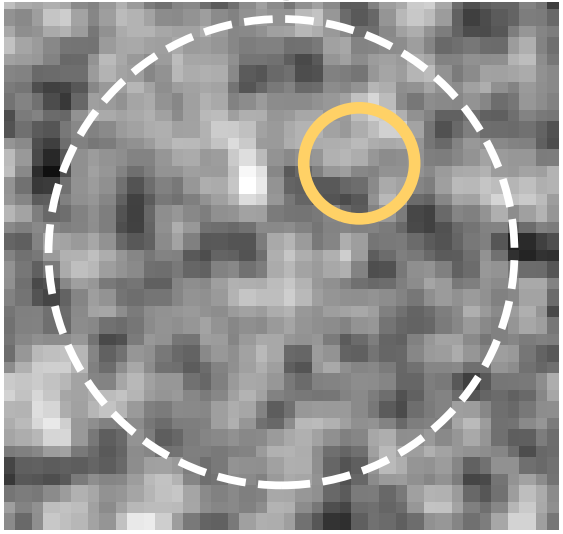
(a) small ROIs look homogeneous

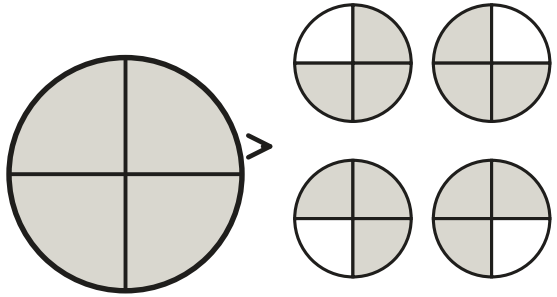


small ROI (123 voxels),
real CT entropy bias:

plug-in **−0.63 bits**

jackknife **−0.01 bits**

(b) block jackknife



ROI split into $B = 8$ blocks

drop one, recompute, combine

cancels the c/N bias for any c

(c) reliability gate

$N/\xi^3 \geq 20$?

ξ: correlation length

yes → **jackknife**

no → **plug-in + flag**

outside the $1/N$ regime
no estimator can recover